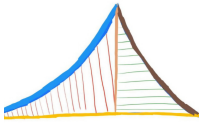
Math for EE



G. V. V. Sharma

About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. CBSE-2020-CLASS-10

A. Formulae

1.1.1 Condition for the quadratic equation to have equal roots

$$b^2 - 4ac = 0 \tag{1.1.1.1}$$

1.1.2 $\mathbf{x} = \begin{bmatrix} x_2 - x_1 \\ y_2 - y_1 \end{bmatrix}$, d is the distance between points

$$d = \sqrt{\mathbf{x}^T\mathbf{x}} \tag{1.1.2.1}$$

1.1.3

$$P(x) = Q(x)Y(x) + R(x) \tag{1.1.3.1}$$

1.1.4 from Basic proportionality theorem

$$\frac{AD}{DB} = \frac{AE}{EC} \tag{1.1.4.1}$$

1.1.5 equation of line ,which has a as x-intercept and b as y-intercept

$$\frac{x}{a} + \frac{y}{b} = 1 \tag{1.1.5.1}$$

B. Problems

1.2.1 The value(s) of k for which the quadratic equation $2x^2 + kx + 2 = 0$ has equal roots, is

a) 4 b) ± 4 c) -4 d) 0

Solution:
For a quadratic equation $ax^2 + bx + c = 0$,condition to have equal roots from (1.1.1.1)

$$b^2 - 4ac = 0 \tag{1.2.1.1}$$

$k^2 - 16 = 0 \implies k = \pm 4$

10/codes/q1.py

1.2.2 which of the following is not an A.P?

a) -1.2,0.8,2.8 ...
b) $3, 3+\sqrt{2}, 3+2\sqrt{2}, ...$
c) $\frac{4}{3}, \frac{7}{3}, \frac{9}{3}, \frac{12}{3}...$
d) $\frac{-1}{5}, \frac{-2}{5}, \frac{-3}{5}....$

Solution:
a.$a_2 - a_1 = 2, a_3 - a_2 = 2$, so,they form an A.P with common difference 2
b.$a_2 - a_1 = \sqrt{2}, a_3 - a_2 = \sqrt{2}$, they form an A.P with common difference $\sqrt{2}$
c.$a_3 - a_2 = \frac{2}{3}, a_4 - a_3 = 1$, no common difference ,so they doesn't form an A.P

1.2.3 The distance between the points $(m, -n)$ and $(-m, n)$ is

a) $\sqrt{m^2+n^2}$ b) $m+n$ c) $2\sqrt{m^2+n^2}$ d) $\sqrt{2m^2+2n^2}$

Solution:
from (1.1.2.1)
distance between two points $(x_1, y_1); (x_2, y_2)$ is

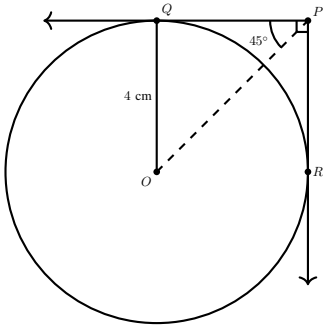
$$d = \sqrt{\mathbf{x}^T\mathbf{x}} \tag{1.2.3.1}$$

$\mathbf{x} = \begin{bmatrix} 2m \\ -2n \end{bmatrix}$
distance $d = \sqrt{4m^2+4n^2} \implies 2\sqrt{m^2+n^2}$

10/codes/q4.c

1.2.4 In Figure-1, from an external point P, two tangents PQ and PR are drawn to a circle of radius 4 cm with centre O. If $\angle QPR = 90°$, then length of PQ is

a) 3 cm b) 4 cm c) 2 cm d) $2\sqrt{2}$cm



Solution: The line joining the centre to the point of intersection bisects the angle between the tangents.
so $\angle OPQ = \frac{\pi}{4}$ $tan(\frac{\pi}{4}) = \frac{OP}{PQ}$
$PQ = OP \implies PQ = 4cm$

1.2.5 On dividing a polynomial $p(x)$ by x^2-4, quotient and remainder are found to be x and 3 respectively. The polynomial $p(x)$ is

a) $3x^2+x-12$ b) x^3-4x+3 c) x^2+3x-4 d) x^3-4x-3

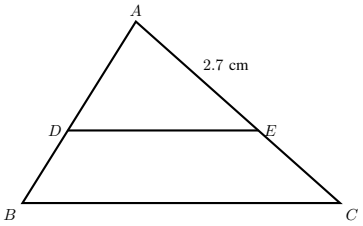
Solution: from (1.1.3.1)

$$P(x) = Q(x)Y(x) + R(x) \tag{1.2.5.1}$$

$P(x), Y(x) = x^2-4, Q(x) = x, R(x) = 3$
$P(x) = x^3-4x+3$

1.2.6 In Figure-2, $DE \parallel BC$. If $\frac{AD}{DB} = \frac{3}{2}$ and $AE = 2 \cdot 7$ cm, then EC is equal to

a) $2 \cdot 0$ cm b) $1 \cdot 8$ cm c) $4 \cdot 0$ cm d) $2 \cdot 7$ cm



Solution:
Basic Proportionality theorem : If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then it divides those two sides in the same ratio.
form (1.1.4.1)
According to the theorem , $\frac{AD}{DB} = \frac{AE}{EC}$
$\frac{2}{3} = \frac{EC}{AE} \implies EC = 1.8cm$

1.2.7 The point on the x-axis which is equidistant from $(-4, 0)$ and $(10, 0)$ is

a) $(7, 0)$ b) $(5, 0)$ c) $(0, 0)$ d) $(3, 0)$

Solution:
let ,the point be P(x,0), Q(10,0),R(-4,0) . Distance between $|PQ| = |PR|$
$|x - 10| = |x + 4| \implies x = 3$

1.2.8 The centre of a circle whose end points of a diameter are $(-6, 3)$ and $(6, 4)$ is

a) $(8, -1)$ b) $(4, 7)$ c) $\left(0, \frac{7}{2}\right)$ d) $\left(4, \frac{7}{2}\right)$

Solution:
centre $\implies$ midpoint of end points of diameter
centre $O(x, y) = (\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}) \implies O(x, y) = (0, \frac{7}{2})$

1.2.9 The pair of linear equations $\frac{3x}{2} + \frac{5y}{3} = 7$ and $9x + 10y = 14$ is

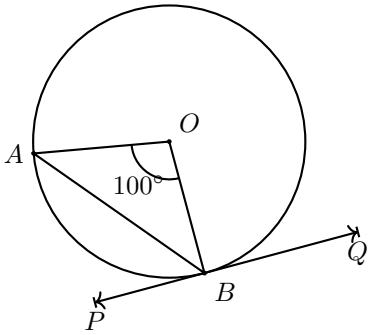
a) consistent
b) inconsistent
c) consistent with one solution
d) consistent with many solutions

Solution:
$9x + 10y = 42$ and $9x + 10y = 14$, they are parallel lines , so lines are inconsistent

10/codes/q9.py

1.2.10 In Figure-3, PQ is tangent to the circle with centre at O, at the point B. If $\angle AOB = 100°$, then $\angle ABP$ is equal to

a) $50°$ b) $40°$ c) $60°$ d) $80°$



Solution:
$\angle OAB = \angle OBA = 40°$
$\angle OBP = 90° \implies \angle ABP = 50°$

1.2.11 Simplest form of $\frac{1+\tan^2 A}{1+\cot^2 A}$ is
Solution:

$$\frac{1+\frac{sin^2A}{cos^2A}}{1+\frac{cos^2A}{sin^2A}} \implies \frac{sin^2A}{cos^2A}$$

$$= tan^2A$$

1.2.12 If the probability of an event E happening is $0 \cdot 023$, then $P(\overline{E}) =$
solution :

$$P(E) + P(\overline{E}) = 1$$

$$P(\overline{E}) = 1 - P(E) \implies P(\overline{E}) = 0.977$$

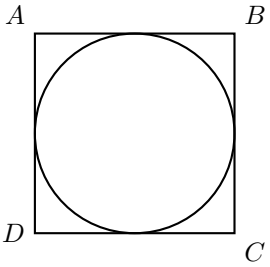
1.2.13 A cone and a cylinder have the same radii but the height of the cone is 3 times that of the cylinder. Find the ratio of their volumes.
Solution:
let V_1 and V_2 be the volumes of cone and cylinder
$r_1 = r_2; h_1 = 3h_2$

$$V_1 = \frac{1}{3}\pi r_1^2 h_1$$

$$V_2 = \pi r_2^2 h_2 \implies \frac{V_1}{V_2} = \frac{h_1}{3h_2} = 1$$

$$\frac{V_1}{V_2} = 1$$

1.2.14 In Figure-4, a quadrilateral ABCD is drawn to circumscribe a circle. Prove that $AB + CD = BC + AD$



Solution:
let the point of intersections of quadrilateral and circle be P,Q,R,S
side AB at P, side BC at Q, side CD at R, side AD at S
$\overline{AP} = \overline{AS}$
$\overline{BP} = \overline{BQ}$,
$\overline{CQ} = \overline{CR}$,
$\overline{DR} = \overline{DS}$
$AP + PB = AB; BQ + QC = BC; CR + RD = CD; DS + AS = AD$

$$AB + CD = AP + PB + CR + RD$$

$$AD + BC = AS + BQ + CQ + DS \implies AP + PB + CR + RD$$

$$AB + CD = AD + BC$$

Hence proved .

1.2.15 Find the sum : (-5)+(-8)+(-11)+ ...+(-230)

Solution: Find the sum of the series:

$$(-5) + (-8) + (-11) + \cdots + (-230)$$

First term (a) = -5
Common difference (d) = $(-8) - (-5) = -3$
Last term (l or a_n) = -230
Using the formula for the n-th term of an A.P., $a_n = a + (n-1)d$:

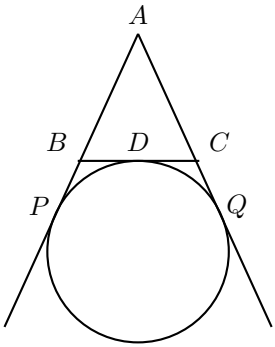
$$\begin{aligned} -230 &= -5 + (n-1)(-3) \\ -230 + 5 &= -3(n-1) \\ -225 &= -3(n-1) \\ n - 1 &= \frac{-225}{-3} \\ n - 1 &= 75 \\ n &= 76 \end{aligned}$$

Using the formula for the sum of an A.P., $S_n = \frac{n}{2}(a+l)$:

$$S_{76} = \frac{76}{2}\left[(-5) + (-230)\right]$$
$$S_{76} = 38 \times (-235)$$
$$S_{76} = -8930$$

The sum of the given A.P. series is -8930

1.2.16 In Figure-5, find the perimeter of ΔABC if AP = 12 cm



Solution:

let $BP = BD = CD = CQ = xcm$ $AP = AQ = 12cm$

$AB = AC = 12 - xcm$

perimeter of $\triangle ABC = 2x + 12 - x + 12 - x$

perimeter = 24cm

1.2.17 Find the mode of the following distribution :

marks	0-10	10-20	20-30	30-40	40-50	50-60
number of students	4	6	7	12	5	6

Solution:

modal class : 30-40

l=30

modal frequency (f_1): 12

$f_0 = 7$

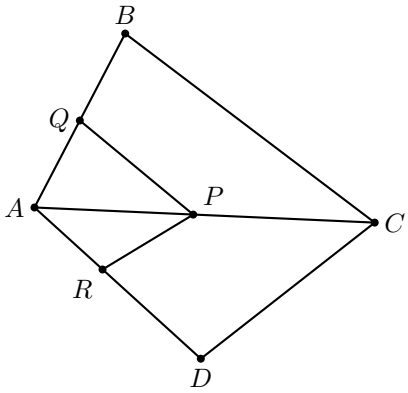
$f_2 = 5$

class size h= 10

mode = $l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) h \implies mode = 30 + \frac{25}{6}$

mode = 34.166

1.2.18 In Figure-7, if $PQ \parallel BC$ and $PR \parallel CD$, prove that $\frac{QB}{AQ} = \frac{DR}{AR}$



solution :
For the $\triangle AQPand\triangle ABC$, given $QP||BC$, by Thale's theorem $\frac{AQ}{QB} = \frac{AP}{PC}$
for the $\triangle ACDand\triangle APR$, given $RP||CD$, by Thale's theorem $\frac{AR}{RD} = \frac{AP}{PC}$

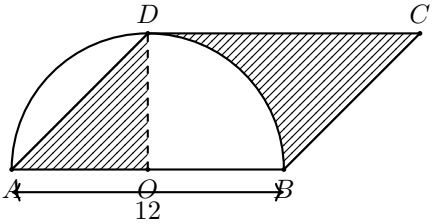
$$\frac{AQ}{QB} = \frac{AR}{RD}$$

1.2.19 Prove that: $(\sin^4\theta - \cos^4\theta + 1)cosec^2\theta = 2$
solution :

$$((sin^2\theta - cos^2\theta)(cos^2\theta + sin^2\theta) + 1)cosec^2\theta$$

$$(1 - cos(2\theta))cosec^2\theta \implies 2sin^2\theta cosec^2\theta = 2$$

1.2.20 In Figure-8, ABCD is a parallelogram. A semicircle with centre O and the diameter AB has been drawn and it passes through D. If $AB = 12$ cm and $OD \perp AB$, then find the area of the shaded region. (Use $\pi = 3 \cdot 14$)



Solution:
$AB = CD = 12cm$ and $AD = BC = 6\sqrt{2}cm$
area of shaded region = area of parallelogram - area of quarter circle
$r = 6cm$, area of quarter circle $= \frac{1}{4}\pi r^2 = 9\pi cm^2$
area of parallelogram $= b.h$; b= length of base
h= perpendicular length from base to opposite side
area of paralellogram = $12 * 6 = 72cm^2$
area of shaded region $= 72 - 9\pi cm^2 = 43.725cm^2$

1.2.21 A fraction becomes $\frac{1}{3}$ when 1 is subtracted from the numerator and it becomes $\frac{1}{4}$

when 8 is added to its denominator. Find the fraction.
Solution:
let fraction be $\frac{x}{y} \implies \frac{x-1}{y} = \frac{1}{3}$

$$3x - y = 3 \tag{1.2.21.1}$$

$\frac{x}{y+8} = \frac{1}{4}$

$$4x - y = 8 \tag{1.2.21.2}$$

so,

$$x = 5, y = 12$$

1.2.22 Find the ratio in which the y-axis divides the line segment joining the points $(6, -4)$ and $(-2, -7)$. Also find the point of intersection.
solution :
equation of line = $3x - 8y = 50$, this line cuts the y- axis at $(0, -\frac{25}{4})$
let $\lambda : 1$ be the ratio , using section formula
$(0, -\frac{25}{4}) = \left(\frac{-2\lambda+6}{\lambda+1}, \frac{-7\lambda-4}{\lambda+1}\right)$
$\lambda = 3$, so ratio is 3:1 and intersection point is $(0, -\frac{25}{4})$

1.2.23 Use Euclid Division Lemma to show that the square of any positive integer is either of the form $3q$ or $3q + 1$ for some integer q.
Solution:
any number can be represented by 3n ,3n+1 or 3n+2 .
$p = (3n)^2 = 9n^2 = 3q$, where q is some integer
$r = (3n + 1)^2 = 9n^2 + 6n + 1 = 3q + 1$, where q is some integer
$r = (3n + 2)^2 = 9n^2 + 12n + 4 = 3(3n^2 + 4n + 1) + 1 = 3q + 1$, where q is some integer
so square of any integer can be shown in the form of 3q,3q+1, whereq is some integer

1.2.24 Sum of the areas of two squares is 544 m^2. If the difference of their perimeters is 32 m, find the sides of the two squares
Solution:
let a , b be side of each square ,sum of area of two squares =$544m^2$
$a^2 + b^2 = 544m^2$
difference of perimeter = 32m
4(a-b)=32m , a-b=8m
$a^2 - 8a - 240 = 0 \implies a = 20m; b = 12m$

1.2.25 A motorboat whose speed is 18 km/h in still water takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream.
Solution:
let v be speed of stream , d=24km
$t = \frac{d}{u}$

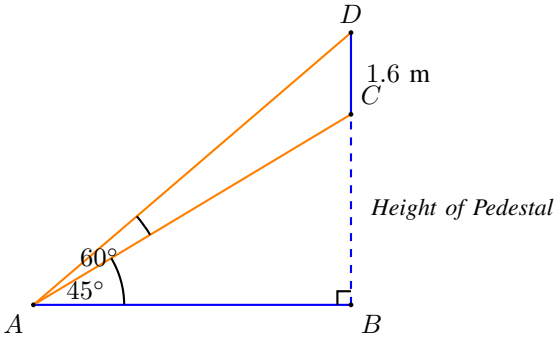
Formula :

$$\frac{d}{18-v} = 1 + \frac{d}{18+v} \tag{1.2.25.1}$$

$$48v = 324 - v^2 \implies v = 6km/h$$

1.2.26 A statue 1.6 m tall, stands on the top of a pedestal. From a point on the ground, the angle of elevation of the top of the statue is $60°$ and from the same point the angle of elevation of the top of the pedestal is $45°$. Find the height of the pedestal. (Use $\sqrt{3} = 1 \cdot 73$)

solution :



let distance from point to lower point of pedestal be x ;
$tan45° = \frac{BC}{AB} = BC = AB = x$
$tan60° = \frac{BD}{AB} = \sqrt{3} \implies \frac{x+1.6}{x} = \sqrt{3}$
height of pedestal =x=$\frac{1.6}{\sqrt{3}-1} = 0.8(\sqrt{3}+1)$

1.2.27 Show that the points $(7, 10)$, $(-2, 5)$ and $(3, -4)$ are vertices of an isosceles right triangle.

Solution:

Let the points be $A(7, 10)$, $B(-2, 5)$, and $C(3, -4)$.
Distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

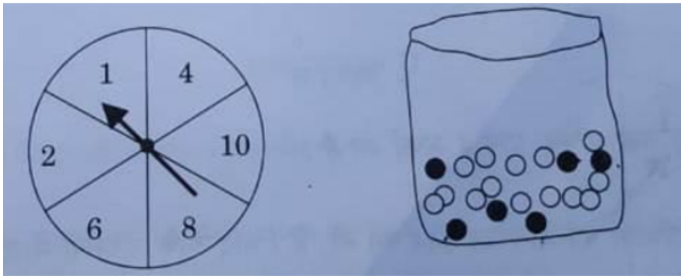
$$AB^2 = (-2-7)^2 + (5-10)^2 = (-9)^2 + (-5)^2 = 81 + 25 = 106$$
$$BC^2 = (3-(-2))^2 + (-4-5)^2 = (5)^2 + (-9)^2 = 25 + 81 = 106$$
$$AC^2 = (3-7)^2 + (-4-10)^2 = (-4)^2 + (-14)^2 = 16 + 196 = 212$$

Since $AB^2 = BC^2 = 106$, we have $AB = BC = \sqrt{106}$.
Because two sides are equal, ΔABC is an isosceles triangle.
3. Verify Right-Angled Property Using the converse of Pythagoras' theorem:

$$AB^2 + BC^2 = 106 + 106 = 212 = AC^2$$

Since $AB^2 + BC^2 = AC^2$, ΔABC is a right-angled triangle at B.
Thus, the given points are the vertices of an isosceles right triangle.

1.2.28 A game in a booth at a Diwali Fair involves using a spinner first. Then, if the spinner stops on an even number, the player is allowed to pick a marble from a bag. The spinner and the marbles in the bag are represented in Fig . Prizes are given, when a black marble is picked. Shweta plays the game once.
A. What is the probability that she will be allowed to pick a marble from the bag?
B.Suppose she is allowed to pick a marble from the bag, what is the probability of getting a prize, when it is given that the bag contains 20 balls out of which 6 are black?



Solution:
let E_1 be the event of getting an even number,and Z be the event of getting a black marble.
A.Favourable outcomes = 5 , total outcomes = 6

$$P(E_1) = \frac{5}{6}$$

B.Since both are independent events $\implies P(E_1 \cap Z) = P(E_1) \cdot P(Z)$
$P(Z|E_1) = \frac{P(Z \cap E_1)}{P(E_1)}$
$P(Z) = \frac{6}{20} \implies P(Z) = \frac{3}{10}$

$$P(Z|E-1) = P(Z) \implies P(Z|E_1) = \frac{3}{10}$$

1.2.29 Obtain other zeroes of the polynomial $p(x) = 2x^4 - x^3 - 11x^2 + 5x + 5$ if two of its zeroes are $\sqrt{5}$ and $-\sqrt{5}$ [Q.38]
Solution:
let x,y be the other two solutions ;
sum of solutions = $-\frac{b}{a} = \frac{1}{2}$
$x + y = \frac{1}{2}$ and product of solutions = $\frac{e}{a} \implies -5xy = \frac{5}{2}$
$xy = -\frac{1}{2} \implies$ x,y are the roots of $2x^2 - x - 1 = 0$
1 and $-\frac{1}{2}$ are the other two roots of the equation $2x^4 - x^3 - 11x^2 + 5x + 5$

10/codes/q29.py

What minimum must be added to $2x^3 - 3x^2 + 6x + 7$ so that the resulting polynomial will be divisible by $x^2 - 4x + 8$?
solution :

$$P(x) = Q(x)Y(x) + R(x) \implies p(x) = 2x^3 - 3x^2 + 6x + 7 \tag{1.2.30.1}$$

$Y(x) = x^2 - 4x + 8 \implies R(x) = 10x - 33$
$P(x) - R(x) = 2x^3 - 3x^2 - 4x + 40$ is divisible by $x^2 - 4x + 8$

1.2.30 The distribution given below shows the number of wickets taken by bowlers in one-day cricket matches. Find the mean and the median of the number of wickets taken.

number of wickets	20-60	60-100	100-140	140-180	180-220	220-260
number of persons	7	5	16	12	2	3

Solution:

Class Interval	**Frequency** (f_i)	**Mid-point** (x_i)	$u_i = \frac{x_i - A}{h}$	$f_i u_i$	cf
20–60	7	40	-2	-14	7
60–100	5	80	-1	-5	12
100–140	16	120	0	0	28
140–180	12	160	1	12	40
180–220	2	200	2	4	42
220–260	3	240	3	9	45
Total	$N = \sum f_i = 45$			$\sum f_i u_i = 6$	

Using the Step-Deviation Formula:

$$\text{Mean } (\bar{x}) = A + \left(\frac{\sum f_i u_i}{N}\right) \times h$$

Substitute $A = 120$, $\sum f_i u_i = 6$, $N = 45$, and $h = 40$:

$$\bar{x} = 120 + \left(\frac{6}{45}\right) \times 40$$

$$\bar{x} = 120 + \left(\frac{2}{15}\right) \times 40 = 120 + \frac{16}{3} = 120 + 5.33 = 125.33$$

a) Find $\frac{N}{2}$:

$$\frac{N}{2} = \frac{45}{2} = 22.5$$

b) The cumulative frequency just greater than 22.5 is 28, which corresponds to the median class **100–140**.

c) Values for the formula:

- Lower limit of median class (l) = 100
- Cumulative frequency of preceding class (cf) = 12
- Frequency of median class (f) = 16
- Class width (h) = 40

Apply the Median formula:

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f}\right) \times h \tag{1.2.31.1}$$

$$\text{Median} = 100 + \left(\frac{22.5 - 12}{16}\right) \times 40 \tag{1.2.31.2}$$

$$\text{Median} = 100 + \left(\frac{10.5}{16}\right) \times 40 = 100 + \frac{10.5 \times 5}{2} = 100 + \frac{52.5}{2} = 100 + 26.25 = 126.25$$

Mean: 125.33 wickets
Median: 126.25 wickets

1.2.32 In a cylindrical vessel of radius 10 cm, containing some water, 9000 small spherical balls are dropped which are completely immersed in water which raises the water level. If each spherical ball is of radius $0 \cdot 5$ cm, then find the rise in the level of water in the vessel.

Solution:

Radius of the cylindrical vessel (R) = 10 cm
Number of spherical balls (n) = 9000
Radius of each spherical ball (r) = 0.5 cm = $\frac{1}{2}$ cm
Let the rise in the water level be h cm.

Formulae

a) Volume of a sphere = $\frac{4}{3}\pi r^3$
b) Volume of a cylinder = $\pi R^2 h$

The volume of water raised in the cylindrical vessel is equal to the total volume of 9000 spherical balls dropped into it: Volume of raised water in cylinder $=$ 9000 $\times$ Volume of one spherical ball
Substitute the respective geometric formulae: $\pi R^2 h = 9000 \times \left(\frac{4}{3}\pi r^3\right)$
Cancel π from both sides:

$$R^2 h = 9000 \times \frac{4}{3} \times r^3$$

Substitute $R = 10$ and $r = 0.5$:

$$(10)^2 \times h = 9000 \times \frac{4}{3} \times (0.5)^3$$

Simplify the terms:

$$100 \times h = 3000 \times 4 \times 0.125$$

$$100 \times h = 12000 \times 0.125$$

$$100 \times h = 1500$$

Solve for h:

$$h = \frac{1500}{100} = 15 \text{ cm}$$

Answer

The rise in the level of water in the vessel is 15 **cm**.

1.2.33 If a line is drawn parallel to one side of a triangle to intersect the other two sides at distinct points, prove that the other two sides are divided in the same ratio

Solution:

1. Given: In ΔABC, a line DE is drawn parallel to side BC ($DE \parallel BC$), intersecting AB at D and AC at E.

2. To Prove:

$$\frac{AD}{DB} = \frac{AE}{EC} \tag{1.2.33.1}$$

Join B to E and C to D.
Draw $DM \perp AC$ (perpendicular from D to AC).
Draw $EN \perp AB$ (perpendicular from E to AB)
3. Step-by-Step Derivation: 1. Calculate the area of ΔADE considering AD as the base:

$$\text{Area}(\Delta ADE) = \frac{1}{2} \times \text{base} \times \text{height} = \frac{1}{2} \times AD \times EN \tag{1.2.33.2}$$

2. Calculate the area of ΔBDE considering BD as the base:

$$\text{Area}(\Delta BDE) = \frac{1}{2} \times BD \times EN \tag{1.2.33.3}$$

3. Divide :

$$\frac{\text{Area}(\Delta ADE)}{\text{Area}(\Delta BDE)} = \frac{\frac{1}{2} \times AD \times EN}{\frac{1}{2} \times BD \times EN} = \frac{AD}{BD} \tag{1.2.33.4}$$

4. Similarly, calculate the area of ΔADE with base AE and height DM, and ΔDEC with base EC and height DM:

$$\frac{\text{Area}(\Delta ADE)}{\text{Area}(\Delta DEC)} = \frac{\frac{1}{2} \times AE \times DM}{\frac{1}{2} \times EC \times DM} = \frac{AE}{EC} \tag{1.2.33.5}$$

5. Relate the areas of ΔBDE and ΔDEC:
Both ΔBDE and ΔDEC lie on the same base DE and between the same parallel lines DE and BC.
Since triangles on the same base and between the same parallels are equal in area:

$$\text{Area}(\Delta BDE) = \text{Area}(\Delta DEC) \quad \text{— (Equation 3)} \tag{1.2.33.6}$$

- Comparing Equations (1), (2), and (3), their left-hand sides are equal because they have identical numerators (Area(ΔADE)) and equal denominators (Area(ΔBDE) = Area(ΔDEC)).
- Therefore, their right-hand sides must be equal:

$$\frac{AD}{DB} = \frac{AE}{EC}$$

2. CBSE-2020-CLASS-12

A. Formulae

FORMULA SHEET

2.1.1 area of a triangle OAB

$$Ar(\triangle OAB) = \frac{1}{2}|\left(OA\right) \times \left(OB\right)| \tag{2.1.1.1}$$

2.1.2

$$sin^{-1}x + cos^{-1}x = \frac{\pi}{2} \tag{2.1.2.1}$$

2.1.3 angle between vectors

$$cos\theta = \left(\frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{u}||\mathbf{v}|}\right) \tag{2.1.3.1}$$

2.1.4 $A \epsilon R^{n\times n}$

$$|A| = |kI| \implies |A| = k^n|I| \tag{2.1.4.1}$$

2.1.5

$$cos^{-1}(-x) = \pi - cos^{-1}x \tag{2.1.5.1}$$

2.1.6 Distance between parallel planes $ax + by + cz = d_1$ and $ax + by + cz = d_2$

$$d = \left|\frac{d_2 - d_1}{\sqrt{a^2+b^2+c^2}}\right| \tag{2.1.6.1}$$

2.1.7 King's property

$$\int_a^b f(x)dx = \int_a^b f(a+b-x)dx \tag{2.1.7.1}$$

2.1.8 Condition for a function to be continuous at x=a

$$f(a^-) = f(a^+) \tag{2.1.8.1}$$

2.1.9 y is a function of x

$$\frac{dy}{dx} + py = q \tag{2.1.9.1}$$

$$\implies ye^{\int pdx} = \int qe^{\int pdx}dx \tag{2.1.9.2}$$

2.1.10 Differentiation of a function$f(x)$ with respect to other function $g(x)$

$$\frac{d(f(x))}{d(g(x))} = \frac{f'(x)}{g'(x)} \tag{2.1.10.1}$$

2.1.11 Let α, β, γ be the angles made by vector with x,y and z axes respectively

$$cos^2\alpha + cos^2\beta + cos^2\gamma = 1 \tag{2.1.11.1}$$

$$\mathbf{a} \cdot \mathbf{b} = |a||b|cos\theta \tag{2.1.11.2}$$

2.1.12 Characteristic polynomial of a matrix A

$$|A - \lambda I| = 0 \tag{2.1.12.1}$$

2.1.13 Integration by parts

$$\int f(x)g(x)dx = f(x)\int g(x)dx - \int f'(x)\left(\int g(x)dx\right)dx \tag{2.1.13.1}$$

2.1.14

$$2sin^{-1}x = sin^{-1}(2x\sqrt{1-x^2}) \tag{2.1.14.1}$$

2.1.15 Inverse of a matrix A

$$A^{-1} = \frac{1}{|A|}adj(A) \tag{2.1.15.1}$$

2.1.16 Condition for minimum surface area

$$S = \pi r^2 + 2\pi rh \implies \frac{dS}{dr} = 0 \tag{2.1.16.1}$$

$$\implies \frac{d^2S}{dr^2} > 0 \tag{2.1.16.2}$$

2.1.17

$$|\mathbf{a} \quad \mathbf{b} \quad \mathbf{c_1} + \mathbf{c_2}| = |\mathbf{a} \quad \mathbf{b} \quad \mathbf{c_1}| + |\mathbf{a} \quad \mathbf{b} \quad \mathbf{c_2}| \tag{2.1.17.1}$$

2.1.18 Equation of a conic

$$\mathrm{g}\left(\mathbf{x}\right) = \mathbf{x}^\top \mathbf{V}\mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \tag{2.1.18.1}$$

2.1.19 pair of straight lines for the intersection of two conics

$$\mathbf{x}^\top \left(\mathbf{V}_1 + \mu\mathbf{V}_2\right)\mathbf{x} + 2\left(\mathbf{u}_1 + \mu\mathbf{u}_2\right)^\top \mathbf{x} + \left(f_1 + \mu f_2\right) = 0 \tag{2.1.19.1}$$

2.1.20 Condition , to get equation of straight line

$$\begin{vmatrix} \mathbf{V}_1 + \mu\mathbf{V}_2 & \mathbf{u}_1 + \mu\mathbf{u}_2 \\ \left(\mathbf{u}_1 + \mu\mathbf{u}_2\right)^\top & f_1 + \mu f_2 \end{vmatrix} = 0 \tag{2.1.20.1}$$

2.1.21 Bayes' theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \tag{2.1.21.1}$$

B. Problems

2.2.1 The area of a triangle formed by vertices O, A and B, where $\left(OA\right)$ =$\mathbf{i} + 2\mathbf{j} + 3\mathbf{k}$ and $\left(OB\right)$ =$-3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ is

a) $3\sqrt{5}$ sq. units b) $5\sqrt{5}$ sq. units c) $6\sqrt{5}$ sq. units d) 4 sq. units

Solution:
For two given vectors (OA) and (OB),
the area of the triangle OAB is given by :
from (2.1.1.1)

$$Ar(\triangle OAB) = \frac{1}{2}\left|((OA) \times (OB))\right| \tag{2.2.1.1}$$

so, area of $\triangle OAB$ is

$$\frac{1}{2}\begin{pmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 3 \\ -3 & -2 & 1 \end{pmatrix}$$

$$|4\mathbf{i} - 5\mathbf{j} + 2\mathbf{k}| = \sqrt{4^2 + (-5)^2 + 2^2}$$

$$Ar(\triangle OAB) = 3\sqrt{5} sq.units$$

.

2.2.2 If $cos(sin^{-1}\frac{2}{\sqrt{5}} + cos^{-1}x) = 0$, then x is equal to

a) $\frac{1}{\sqrt{5}}$ b) $-\frac{2}{\sqrt{5}}$ c) $\frac{2}{\sqrt{5}}$ d) 1

Solution:
from (2.1.2.1)

$$cos^{-1}x + sin^{-1}x = \frac{\pi}{2} \tag{2.2.2.1}$$

$cos(\frac{\pi}{2}) = 0$ so,

$$cos^{-1}\frac{2}{\sqrt{5}} + sin^{-1}x = \frac{\pi}{2}$$

$$x = \frac{2}{\sqrt{5}}$$

.

2.2.3 The interval in which the function f given by $f(x) = x^2e^{-x}$ is strictly increasing is

a) $(-\infty, \infty)$ b) $(-\infty, 0)$ c) $(2, \infty)$ d) $(0, 2)$

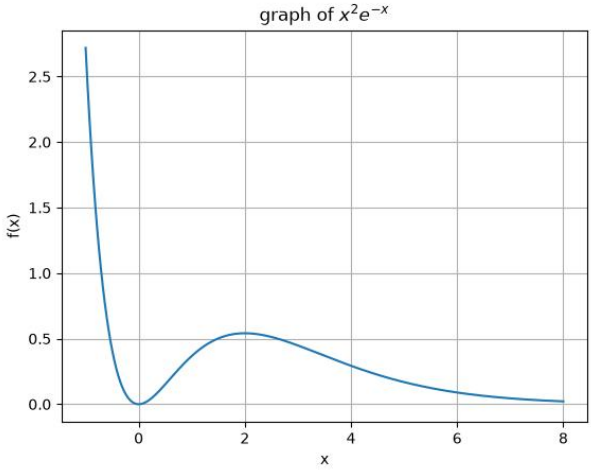
Solution:

$$f(x) = x^2e^{-x}$$

$$f'(x) = 2xe^{-x} - x^2e^{-x} = 0 \tag{2.2.3.1}$$

$2x - x^2 > 0$,for the $f(x)$ to be strictly increasing
so ,

$$x \in (0, 2)$$

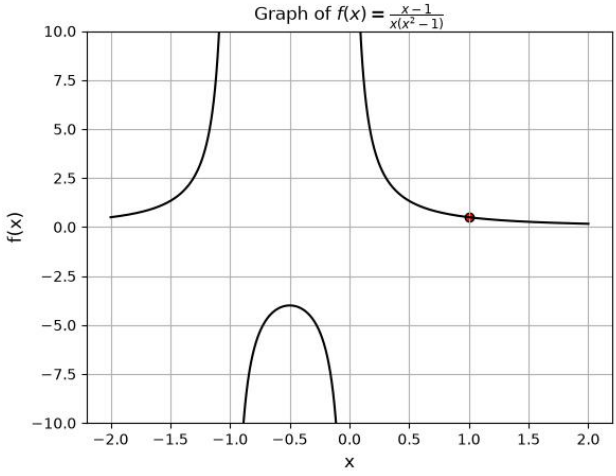


12/codes/q3.py

2.2.4 The function $f(x) = \frac{x-1}{x(x^2-1)}$ is discontinuous at

a) exactly one point
b) exactly two points
c) no point
d) exactly three points

Solution:
$f(x)$ domain $\mathbf{R} - \{0, 1, -1\}$, so $f(x)$ is discontinuous at $x = 0, 1, -1$.
3 points of discontinuity



12/codes/q4.py

2.2.5 The function $f : \mathbf{R} \to [-1, 1]$ defined by $f(x) = cos\ x$ is

a) both one-one and onto
b) not one-one, but onto
c) one-one, but not onto
d) neither one-one, nor onto

Solution:
Range of $cosx$ is [-1,1] , so the function $f(x)$ is surjective or onto $cosx$ in $\mathbf{R}$ is not one-one.

12/codes/q5.py

The coordinates of the foot of the perpendicular drawn from the point $(2, -3, 4)$ on the y-axis is

2.2.6 a) $(2, 3, 4)$ b) $(-2, -3, -4)$ c) $(0, -3, 0)$ d) $(2, 0, 4)$

Solution:
foot of perpendicular of the point (x_1, y_1, z_1) on the y-axis is

$$(0, y_1, 0) \implies (0, -3, 0)$$

2.2.7 The relation R in the set 1,2,3 given by $R = \{(1, 2), (2, 1), (1, 1)\}$is

a) symmetric and transitive ,but not reflexive
b) reflexive and symmetric ,but not transitive
c) symmetric ,but neither reflexive nor transitive
d) an equivalence relation

Solution: Relation R is not reflexive ,since $(2, 2), (3, 3) \notin R$
Relation R is not transitive $(2, 1)\epsilon R, (1, 2)\epsilon R$ but $(2, 2) \notin R$
R is symmetric.

2.2.8 The angle between the vectors $\mathbf{i} - \mathbf{j}$ and $\mathbf{j} - \mathbf{k}$ is

a) $-\frac{\pi}{3}$ b) 0 c) $\frac{\pi}{3}$ d) $\frac{2\pi}{3}$

Solution:
angle between two vectors $a_1\mathbf{i} + b_1\mathbf{j} + c_1\mathbf{k}$ and $a_2\mathbf{i} + b_2\mathbf{j} + c_2\mathbf{k}$ is
from (2.1.3.1)

$$cos\theta = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}\right) \tag{2.2.8.1}$$

from (2.1.5.1)

$$cos\theta = \left(\frac{-1}{2}\right)$$

so $\theta = \frac{2\pi}{3}$

2.2.9 If A is a non-singular square matrix of order 3 such that $A^2 = 3A$ then value of $|A|$ is

a) -3 b) 3 c) 9 d) 27

Solution:
from (2.1.4.1)

$$|A^2| = |3A|, |A| \neq 0 \tag{2.2.9.1}$$

$$|A| = 27$$

12/codes/q9.py

If $|\mathbf{a}| = 4$ and $-3 \leq \lambda \leq 2$, then $|\lambda\mathbf{a}|$ lies in

2.2.10 a) $[0, 12]$ b) $[2, 3]$ c) $[8, 12]$ d) $[-12, 8]$

Solution:
$-3 \leq \lambda \leq 2$,then $0 \leq |\lambda| \leq 3$
$|\lambda\mathbf{a}|$, $4|\lambda| \in [0, 12]$
so,

$$|\lambda\mathbf{a}| \in [0, 12]$$

2.2.11 If the radius of the circle is increasing at the rate of 0.5 cm/s, then the rate of increase of its circumference is
Solution:
$\frac{dr}{dt} = 0.5cm/s$, circumference $c = 2\pi r$
$\frac{dc}{dt} = 2\pi\frac{dr}{dt}$, $\frac{dc}{dt} = \pi cm/s$

2.2.12 If $\begin{vmatrix} 2x & -9 \\ -2 & x \end{vmatrix} = \begin{vmatrix} -4 & 8 \\ 1 & -2 \end{vmatrix}$ then value of x is
Solution:
$2x^2 - 18 = 0$, $x = \pm 3$

12/codes/q12.py

The corner points of the feasible region of an LPP are $(0, 0)$ $(0, 8)$, $(2, 7)$, $(5, 4)$ and $(6, 0)$ The maximum profit $P = 3x + 2y$ occurs at the point
Solution:
To find the point at which the maximum value of the objective function $P = 3x + 2y$ occurs, we evaluate P at each of the given corner points[cite: 4]:

2.2.13 a) At $(0, 0)$:

$$P = 3(0) + 2(0) = 0$$

b) At $(5,4)$:

$$P = 3(5) + 2(4) = 15 + 8 = 23 \quad \text{(Maximum)}$$

similarly ,we evaluate it at all corners and it is maximum at $(5,4)$
Comparing all the values, the maximum value of P is 23, which occurs at the point $(5,4)$
The maximum profit occurs at the point $(5,4)$

2.2.14 The distance between parallel planes $2x + y - 2z - 6 = 0$ and $4x + 2y - 4z = 0$ is units

Solution:
$4x + 2y - 4z = 12$ and $4x + 2y - 4z = 0$
distance between parallel planes $ax + by + cz = d_1$ and $ax + by + cz = d_2$ is
$\mathbf{u} = 4\mathbf{i} + 2\mathbf{j} - 4\mathbf{k}$
from (2.1.6.1)

$$D = \left| \frac{d_1 - d_2}{||u||} \right| \tag{2.2.14.1}$$

$$D = \left| \frac{12}{\sqrt{4^2 + 2^2 + 4^2}} \right|$$

, D =2 .

or

2.2.15 If P(1,0,-3) is the foot of the perpendicular from origin to the plane , then the cartesian equation of the plane is

Solution: $(OP) = \mathbf{i} - 3\mathbf{k}$ is perpendicular to every point in the plane ,so (OP) can be considered as normal vector of the plane
so equation of the plane

$$1(x-1) + 0(y) - 3(z+3) = 0 \implies x - 3z = 9 \tag{2.2.15.1}$$

2.2.16 Evaluate: $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x cos^2 x dx$

Solution:
from (2.1.7.1)

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x cos^2 x dx \tag{2.2.16.1}$$

$$I = \int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \left(\frac{\pi}{2} - \frac{\pi}{2} - x \right) cos^2 x dx$$

I = -I ; I=0.

2.2.17 Find the coordinates of the point where the line $\frac{x-1}{3} = \frac{y+4}{7} = \frac{z+4}{2}$ cuts the xy-plane.

Solution:
z=0 ,

$$\frac{x-1}{3} = \frac{y+4}{7} = 2$$

$$x = 7, y = 10$$

2.2.18 Find the value of k, so that the function $f(x) = \begin{cases} kx^2 + 5 & if x \leq 1 \\ 2 & if x > 1 \end{cases}$ is continuous at $x = 1$

Solution:

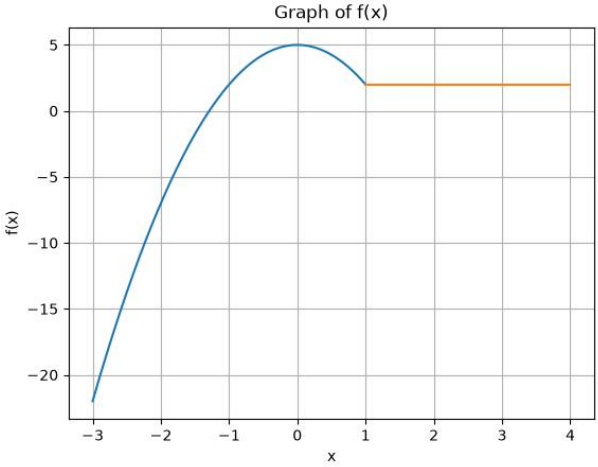
from (2.1.8.1)

$$f(1^-) = f(1^+) \tag{2.2.18.1}$$

$$k + 5 = 2$$

$$k = -3$$

for k=-3,graph of the function would be,



12/codes/q18.py

Find the integrating factor of the differential equation $x\frac{dy}{dx} = 2x^2 + y$

Solution:

from (2.1.9.2)

$$x\frac{dy}{dx} - y = 2x^2 \tag{2.2.19.1}$$

$$\implies \frac{dy}{dx} - \frac{y}{x} = 2x$$

$$ye^{\int -\frac{1}{x}dx} = \int 2xe^{\int -\frac{1}{x}dx}dx$$

integrating factor $e^{\int -\frac{1}{x}dx}$

2.2.19 If $y = f(x^2)$ and $f'(x) = e^{\sqrt{x}}$ then find $\frac{dy}{dx}$

Solution:

$\frac{dy}{dx} = 2xf'(x^2) \implies \frac{dy}{dx} = 2xe^x$

2.2.21 **a.**Find a vector $\mathbf{r}$ equally inclined to the three axes and whose magnitude is $3\sqrt{3}$ units.

Solution:

from (2.1.11.1)

$\alpha = \beta = \gamma = \cos^{-1}(\pm\frac{1}{3})$

$\overrightarrow{r} = 3\sqrt{3}\left(\pm\frac{1}{\sqrt{3}}\hat{i} \pm \frac{1}{\sqrt{3}}\hat{j} \pm \frac{1}{\sqrt{3}}\hat{k}\right)$

or

b.Find the angle between unit vectors $\mathbf{a}$ and $\mathbf{b}$ so that $\sqrt{3}\mathbf{a} - \mathbf{b}$ is also a unit vector.

Solution:

$|\sqrt{3}\mathbf{a} - \mathbf{b}| = \sqrt{3|a|^2 + |b|^2 + 2\sqrt{3}\,(\mathbf{a}\cdot\mathbf{b})}$

$1 = \sqrt{3 + 1 + 2\sqrt{3}\,(\mathbf{a}\cdot\mathbf{b})}$

$\mathbf{a}\cdot\mathbf{b} = -\frac{\sqrt{3}}{2}$

from (2.1.11.2)

$$cos\theta = -\frac{\sqrt{3}}{2} \tag{2.2.21.1}$$

$\implies \theta = \frac{5\pi}{6}$

2.2.22 If $A = \begin{pmatrix} -3 & 2 \\ 1 & -1 \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ find scalar k so that $A^2 + I = kA$.

Solution:

$|A - \lambda I| = 0 \implies \begin{vmatrix} -3-\lambda & 2 \\ 1 & -1-\lambda \end{vmatrix} = 0$

$\lambda^2 + 4\lambda + 1 = 0 \implies A^2 + I = -4A \implies k = -4$

2.2.23 **a.**If $f(x) = \sqrt{\frac{secx-1}{secx+1}}$, find $f'(\frac{\pi}{3})$

Solution:

$f(x) = \sqrt{\frac{1-cosx}{1+cosx}} \implies f(x) = tan\frac{x}{2}$

$f'(x) = \frac{1}{2}sec^2\frac{x}{2} \implies f'(\frac{\pi}{3}) = \frac{2}{3}$

or

b. Find $f'(x)$ if $f(x) = (tan\ x)^{tan\ x}$

Solution:

from (2.1.13.1)

$$f'(x) = tanx(tanx)^{tanx-1}(sec^2x) + tanx^{tanx}ln(tanx)sec^2x \tag{2.2.23.1}$$

$f'(x) = tanx^{tanx}sec^2x\,(1 + ln(tanx))$

2.2.24 Find $\int \frac{tan^3x}{cos^3x}dx$

Solution:

cosx=t ; sinxdx=-dt

$\int \frac{sin^3x}{cos^6x}dx \implies \int \frac{cos^2x-1}{cos^6x}(-sinxdx) \implies \int \frac{t^2-1}{t^6}dt \implies \int \frac{1}{t^4}dt - \frac{1}{t^6}dt$

$\frac{1}{5}sec^5x - \frac{1}{3}sec^3x$

2.2.25 Show that the plane $x - 5y - 2z = 1$ contains the line $\frac{x-5}{3} = y = 2 - z$.

Solution:

$\frac{x-5}{3} = y = 2 - z = k \implies x = 3k + 5, y = k, z = 2 - k$

substituting them in the plane equation gives,

$3k + 5 - 5k - 2(2 - k) - 1 = 0 \implies 0 = 0$, so the line lies in the plane

2.2.26 A fair dice is thrown two times. Find the probability distribution of the number of sixes. Also determine the mean of the number of sixes.

Solution:

X,Y are the outcomes of each die,

$$P(X,Y) = \begin{cases} \frac{25}{36} & X \neq 6, Y \neq 6 \\ \frac{5}{18} & X = 6, Y \neq 6 (or) X \neq 6, Y = 6 \\ \frac{1}{36} & X = 6, Y = 6 \end{cases}$$

mean $\mu = E(x) = \Sigma X \cdot P(X)$

$E(x) = (0 * \frac{25}{26}) + (1 * \frac{5}{18}) + (2 * \frac{1}{36})$

$$E(x) = \frac{1}{3}$$

2.2.27 Solve the following differential equation

$(1 + e^{y/x})dy + e^{y/x}(1 - \frac{y}{x})dx = 0 (x \neq 0)$.

Solution:

Let $\frac{y}{x} = t \implies \frac{dy}{dx} = t + x\frac{dt}{dx}$

$$(1 + e^t)(t + x\frac{dt}{dx}) + e^t(1 - t) = 0$$

$$\frac{e^t + t}{e^t + 1}dt = -\frac{1}{x}dx$$

$$ln(e^t + t) = ln(\frac{1}{x}) \implies e^t + t = \frac{1}{x}$$

$$e^{\frac{y}{x}} + \frac{y}{x} = \frac{1}{x}$$

2.2.28 Evaluate :

$$\int_0^{\frac{\pi}{2}} sin2xtan^{-1}(sinx)dx$$

Solution:

$I = \int_0^{\frac{\pi}{2}} sin2xtan^{-1}(sinx)dx = \int_0^{\frac{\pi}{2}} sin2xtan^{-1}(cosx)dx$

$sinx = t; cosxdx = dt$

$I = 2\int_0^1 ttan^{-1}tdt$

from (2.1.13.1)

$$I = 2\left(\left(\frac{t^2}{2}tan^{-1}t\right)_0^1 - \int_0^1 \frac{t^2}{2(t^2+1)}dt\right) \tag{2.2.28.1}$$

$I = 2\left(\frac{\pi}{8} - \frac{1}{2}\left[t\right]_0^1 + \frac{1}{2}\left[tan^{-1}t\right]_0^1\right)$

$I = \frac{\pi}{2} - 1$

2.2.29 Prove that $tan^{-1}\frac{1}{4} + tan^{-1}\frac{2}{9} = \frac{1}{2}sin^{-1}(\frac{4}{5})$

Solution:

$tan^{-1}\left(\frac{\frac{1}{4}+\frac{2}{9}}{1-\frac{1}{18}}\right)$

$tan^{-1}(\frac{1}{2}) = sin^{-1}(\frac{1}{\sqrt{5}})$

from (2.1.14.1)

$$2sin^{-1}(\frac{1}{\sqrt{5}}) = sin^{-1}(\frac{4}{5})$$

$\implies sin^{-1}(\frac{1}{\sqrt{5}}) = \frac{1}{2}sin^{-1}(\frac{4}{5})$

Hence proved

2.2.30 Find the equation of the plane passing through the points $(1, 0, -2)$, $(3, -1, 0)$ and perpendicular to the plane $2x - y + z = 8$ Also find the distance of the plane thus obtained from the origin.

Solution:

normal vector of plane 1 i.e, $(2, -1, 1)$ lies on plane 2

dr be the normal vector of plane 2

$$dr_1 = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}; dr_2 = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

$$dr = dr_1 \times dr_2 \implies \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$$

$1(x-1) + 2(y) + 0(z+2) = 0 \implies x + 2y = 1$

2.2.31 If $tan^{-1}(\frac{y}{x}) = log\sqrt{x^2+y^2}$, prove that $\frac{dy}{dx} = \frac{x+y}{x-y}$

Solution:

$\frac{y}{x} = t \implies \frac{xdy-y}{x^2} = dt$

$tan^{-1}t = \frac{1}{2}(2logx + log(1+t^2)) \implies \frac{1}{1+t^2}dt = \frac{1}{x} + \frac{tdt}{1+t^2}$

$$\frac{1-t}{1+t^2}dt = \frac{1}{x}$$

$$\frac{x(x-y)}{x^2+y^2}\left(\frac{x\frac{dy}{dx} - y}{x^2}\right) = \frac{1}{x}$$

$$x\frac{dy}{dx} - y = \frac{x^2+y^2}{x-y} \tag{2.2.31.1}$$

$$\frac{dy}{dx} = \frac{x+y}{x-y}$$

or

question :

If $y = e^{a\ cos^{-1}x}, -1 < x < 1$, then show that $(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} - a^2y = 0$

Solution:

$\frac{dy}{dx} = -e^{acos^{-1}x}a(\frac{1}{\sqrt{1-x^2}}) \implies \frac{dy}{dx} = \frac{-ay}{\sqrt{1-x^2}} \implies \sqrt{1-x^2}\frac{dy}{dx} = -ay$

$$(1-x^2)(\frac{dy}{dx})^2 = a^2y^2$$

$$2\frac{d^2y}{dx^2}\frac{dy}{dx}(1-x^2) - 2x\left(\frac{dy}{dx}\right)^2 = a^2y^2$$

$$2\frac{d^2y}{dx^2}(\frac{-ay}{\sqrt{1-x^2}})(1-x^2) - 2x\frac{a^2y^2}{1-x^2} = 2a^2y(\frac{-ay}{\sqrt{1-x^2}}) \tag{2.2.31.2}$$

$$(1-x^2)\frac{d^2y}{dx^2} - x\left(\frac{-ay}{\sqrt{1-x^2}}\right) = a^2y$$

$$(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} - a^2y = 0 \tag{2.2.31.3}$$

2.2.32 Amongst all open (from the top) right circular cylindrical boxes of volume $125\pi\ cm^3$ find the dimensions of the box which has the least surface area.

Solution:

$V = \pi r^2 h \implies h = \frac{125}{r^2}$

The total surface area S

$$S = \pi r^2 + 2\pi rh \tag{2.2.32.1}$$

$$S(r) = \pi r^2 + 2\pi r\left(\frac{125}{r^2}\right) = \pi r^2 + \frac{250\pi}{r} \tag{2.2.32.2}$$

Differentiating with respect to r:

$$\frac{dS}{dr} = 2\pi r - \frac{250\pi}{r^2} \tag{2.2.32.3}$$

from (2.1.16.2)

$$\frac{dS}{dr} = 0$$

:

$$2\pi r - \frac{250\pi}{r^2} = 0 \tag{2.2.32.4}$$

$$2\pi r^3 = 250\pi$$

$$r^3 = 125 \implies r = 5 \text{ cm}$$

Second derivative test:

$$\frac{d^2S}{dr^2} = 2\pi + \frac{500\pi}{r^3} \tag{2.2.32.5}$$

At $r = 5$:

$$\left.\frac{d^2S}{dr^2}\right|_{r=5} = 2\pi + \frac{500\pi}{125} = 2\pi + 4\pi = 6\pi > 0 \tag{2.2.32.6}$$

the surface area is minimum when $r = 5$ cm.
Corresponding height:
$h = \frac{125}{(5)^2} = \frac{125}{25} = 5$ cm
Radius $r = 5$ cm and Height $h = 5$ cm.

2.2.33 Using integration, find the area lying above x-axis and included between the circle $x^2 + y^2 = 8x$ and inside the parabola $y^2 = 4x$

Solution:
intersection of two conics is defined as

$$\mathbf{x}^\top (\mathbf{V}_1 + \mu\mathbf{V}_2)\,\mathbf{x} + 2\,(\mathbf{u}_1 + \mu\mathbf{u}_2)^\top \mathbf{x} + (f_1 + \mu f_2) = 0 \tag{2.2.33.1}$$

(2.1.20.1) represents pair of staright lines if

$$\begin{vmatrix} \mathbf{V}_1 + \mu\mathbf{V}_2 & \mathbf{u}_1 + \mu\mathbf{u}_2 \\ (\mathbf{u}_1 + \mu\mathbf{u}_2)^\top & f_1 + \mu f_2 \end{vmatrix} = 0 \tag{2.2.33.2}$$

for circle ,

$$\mathbf{V}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{u}_1 = \begin{pmatrix} -4 \\ 0 \end{pmatrix}, \quad f_1 = 0 \tag{2.2.33.3}$$

for $y^2 = 4x$

$$\mathbf{V}_2 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} -2 \\ 0 \end{pmatrix}, \quad f_2 = 0 \tag{2.2.33.4}$$

$$\begin{vmatrix} 1 & 0 & -4 - 2\mu \\ 0 & 1 + \mu & 0 \\ -4 - 2\mu & 0 & 0 \end{vmatrix} = 0$$

$$\implies (4 + 2\mu)^2(1 + \mu) = 0$$

substituting the μ values in first equation gives
for $\mu = -2$

$$\mathbf{x}^\top \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{x} = 0$$

$$x = \pm y$$

for $\mu = -1$

$$\mathbf{x}^\top \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{x} + 2 \begin{bmatrix} -2 & 0 \end{bmatrix} \mathbf{x} = 0$$

$$x = 0, x = 4$$

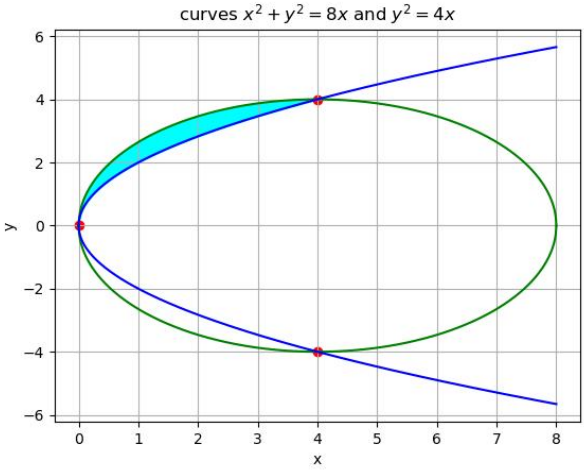
The required area above the x-axis is split into two regions: 1. From $x = 0$ to $x = 4$, bounded by the parabola $y = 2\sqrt{x}$. 2. From $x = 4$ to $x = 8$, bounded by the circle $y = \sqrt{8x - x^2}$.

$$\text{Area} = -\int_0^4 2\sqrt{x}\,dx + \int_4^8 \sqrt{8x - x^2}\,dx$$

$$= -2\left[\frac{x^{3/2}}{3/2}\right]_0^4 + \left[\frac{x-4}{2}\sqrt{8x - x^2} + \frac{16}{2}\sin^{-1}\left(\frac{x-4}{4}\right)\right]_4^8$$

$$= -\frac{4}{3}(4^{3/2}) + \left[0 + 8\sin^{-1}(1)\right] - \left[0 + 8\sin^{-1}(0)\right]$$

$$= -\frac{32}{3} + 8\left(\frac{\pi}{2}\right) = \left(4\pi - \frac{32}{3}\right) \text{ sq. units}$$



12/codes/q33.py

or

Question :

Using the method of integration, find the area of the triangle ABC, coordinates of whose vertices are $A(2, 0)$, $B(4, 5)$ and $C(6, 3)$.

Solution:

Equation of line AB passing through $A(2,0)$ and $B(4,5)$:

$$y-0=\frac{5-0}{4-2}(x-2)$$

$$y=\frac{5}{2}(x-2) \quad \implies y_1=\frac{5}{2}(x-2)$$

Equation of line BC passing through $B(4,5)$ and $C(6,3)$:

$$y-5=\frac{3-5}{6-4}(x-4)$$

$$y-5=\frac{-2}{2}(x-4)$$

$$y-5=-(x-4) \quad \implies y_2=9-x$$

Equation of line AC passing through $A(2,0)$ and $C(6,3)$:

$$y-0=\frac{3-0}{6-2}(x-2)$$

$$y=\frac{3}{4}(x-2) \quad \implies y_3=\frac{3}{4}(x-2)$$

$\text{Ar}(\triangle ABC)=$ Area under $AB\,(x\in[2,4])+$Area under $BC(x\in[4,6])-$Area under AC

$$=\int_2^4\frac{5}{2}(x-2)\,dx+\int_4^6(9-x)\,dx-\int_2^6\frac{3}{4}(x-2)\,dx$$

Evaluating the integrals individually:

$$I_1=\frac{5}{2}\left[\frac{(x-2)^2}{2}\right]_2^4=\frac{5}{2}\left(\frac{4}{2}-0\right)=5$$

$$I_2=\left[9x-\frac{x^2}{2}\right]_4^6=(54-18)-(36-8)=36-28=8$$

$$I_3=\frac{3}{4}\left[\frac{(x-2)^2}{2}\right]_2^6=\frac{3}{4}\left(\frac{16}{2}-0\right)=6$$

Combining the results:

$$\begin{aligned}\text{Area}(\triangle ABC)&=I_1+I_2-I_3\\&=5+8-6\\&=7\text{ sq. units}\end{aligned}$$

area $\implies$ 7 sq. units

2.2.34 If $A=\begin{pmatrix}5&-1&4\\2&3&5\\5&-2&6\end{pmatrix}$ find A^{-1} and use it to solve the following system of equations:
$5x-y+4z=5$

$2x + 3y + 5z = 2$
$5x - 2y + 6z = -1$

Solution: $A = \begin{pmatrix} 5 & -1 & 4 \\ 2 & 3 & 5 \\ 5 & -2 & 6 \end{pmatrix}$ and $B = \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix}$

$$[A \mid B] = \left[\begin{array}{ccc|c} 5 & -1 & 4 & 5 \\ 2 & 3 & 5 & 2 \\ 5 & -2 & 6 & -1 \end{array}\right]$$

Apply $R_2 \to 5R_2 - 2R_1$ and $R_3 \to R_3 - R_1$:

$$\left[\begin{array}{ccc|c} 5 & -1 & 4 & 5 \\ 0 & 17 & 17 & 0 \\ 0 & -1 & 2 & -6 \end{array}\right]$$

Apply $R_2 \to \frac{1}{17}R_2$:

$$\left[\begin{array}{ccc|c} 5 & -1 & 4 & 5 \\ 0 & 1 & 1 & 0 \\ 0 & -1 & 2 & -6 \end{array}\right]$$

Apply $R_3 \to R_3 + R_2$:

$$\left[\begin{array}{ccc|c} 5 & -1 & 4 & 5 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 3 & -6 \end{array}\right] \tag{2.2.34.1}$$

$$3z = -6 \implies z = -2$$

Substitute $z = -2$ into the second row:

$$y + z = 0 \implies y + (-2) = 0 \implies y = 2$$

Substitute $y = 2$ and $z = -2$ into the first row:

$$5x - y + 4z = 5 \implies 5x - 2 + 4(-2) = 5 \implies 5x - 10 = 5 \implies 5x = 15 \implies x = 3$$

$$x = 3, y = 2, z = -2$$

Another solution:

$$\begin{aligned} |A| &= 5(18 - (-10)) - (-1)(12 - 25) + 4(-4 - 15) \\ &= 5(28) + 1(-13) + 4(-19) \\ &= 140 - 13 - 76 = 51 \neq 0 \text{ (non-singular)} \end{aligned}$$

Cofactors of A:

$$C_{11} = 28, \quad C_{12} = 13, \quad C_{13} = -19$$
$$C_{21} = -2, \quad C_{22} = 10, \quad C_{23} = 5$$
$$C_{31} = -17, \quad C_{32} = -17, \quad C_{33} = 17$$

$$\text{Adj}(A) = \begin{pmatrix} 28 & -2 & -17 \\ 13 & 10 & -17 \\ -19 & 5 & 17 \end{pmatrix} \tag{2.2.34.2}$$

From (2.1.15.1)

$$A^{-1} = \frac{1}{|A|}\text{Adj}(A) = \frac{1}{51}\begin{pmatrix} 28 & -2 & -17 \\ 13 & 10 & -17 \\ -19 & 5 & 17 \end{pmatrix} \tag{2.2.34.3}$$

The matrix equation is $AX = B$, where $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ and $B = \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix}$.

$$X = A^{-1}B = \frac{1}{51}\begin{pmatrix} 28(5) - 2(2) - 17(-1) \\ 13(5) + 10(2) - 17(-1) \\ -19(5) + 5(2) + 17(-1) \end{pmatrix}$$
$$= \frac{1}{51}\begin{pmatrix} 140 - 4 + 17 \\ 65 + 20 + 17 \\ -95 + 10 - 17 \end{pmatrix} = \frac{1}{51}\begin{pmatrix} 153 \\ 102 \\ -102 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ -2 \end{pmatrix}$$

12/codes/q34.py

or

.

Question : If x, y, z are different and

$$\begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0$$

then using properties of determinants show that $1 + xyz = 0$

Solution:

Let $\Delta = \begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix}$.

Using the property of determinants that allows splitting a column containing a sum of two terms:

$$\Delta = \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ y & y^2 & y^3 \\ z & z^2 & z^3 \end{vmatrix}$$

Taking out x, y, z common from the $1^{st}, 2^{nd}$, and 3rd rows respectively in the second determinant:

$$\Delta = \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + xyz \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$$

In the first determinant, interchange columns to match the second determinant:

$$\begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} = - \begin{vmatrix} x & 1 & x^2 \\ y & 1 & y^2 \\ z & 1 & z^2 \end{vmatrix} \quad (C_2 \leftrightarrow C_3)$$

$$= (-1)^2 \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} \quad (C_1 \leftrightarrow C_2)$$

$$= \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$$

$$\Delta = \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} + xyz \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$$

$$\Delta = (1 + xyz) \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix}$$

Now, apply row operations to evaluate the determinant:

$$R_2 \to R_2 - R_1$$

$$R_3 \to R_3 - R_1$$

$$\Delta = (1 + xyz) \begin{vmatrix} 1 & x & x^2 \\ 0 & y - x & y^2 - x^2 \\ 0 & z - x & z^2 - x^2 \end{vmatrix}$$

Take $(y - x)$ common from R_2 and $(z - x)$ common from R_3:

$$\Delta = (1 + xyz)(y - x)(z - x) \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & y + x \\ 0 & 1 & z + x \end{vmatrix}$$

$$\Delta = (1 + xyz)(y - x)(z - x)[1 \cdot (z + x - (y + x))]$$

$$= (1 + xyz)(y - x)(z - x)(z - y)$$

$$= (1 + xyz)(x - y)(y - z)(z - x)$$

We are given that $\Delta = 0$:

$$(1 + xyz)(x - y)(y - z)(z - x) = 0$$

Since x, y, z are distinct, $(x - y) \neq 0$, $(y - z) \neq 0$, and $(z - x) \neq 0$.
Therefore:

$$1 + xyz = 0$$

2.2.35 A card from a pack of 52 cards is lost. From the remaining cards of the pack, two cards are drawn randomly one-by-one without replacement and are found to be both kings. Find the probability of the lost card being a king.
Solution:

$$\text{Let } E_1 = \text{Lost card is a king} \implies P(E_1) = \frac{4}{52} = \frac{1}{13}$$

$$\text{Let } E_2 = \text{Lost card is not a king} \implies P(E_2) = \frac{48}{52} = \frac{12}{13}$$

$$\text{Let } A = \text{Two cards drawn from remaining 51 cards are both kings}$$

Using conditional probabilities:

$$P(A|E_1) = \frac{\text{3 kings left}}{\text{51 cards left}} = \frac{3_{C_2}}{51_{C_2}} = \frac{3 \times 2}{51 \times 50} = \frac{6}{2550} = \frac{1}{425}$$

$$P(A|E_2) = \frac{\text{4 kings left}}{\text{51 cards left}} = \frac{4_{C_2}}{51_{C_2}} = \frac{4 \times 3}{51 \times 50} = \frac{12}{2550} = \frac{2}{425}$$

from (**??**)
Using Bayes' Theorem to find $P(E_1|A)$:

$$P(E_1|A) = \frac{P(E_1)P(A|E_1)}{P(E_1)P(A|E_1) + P(E_2)P(A|E_2)}$$

$$= \frac{\frac{1}{13} \times \frac{1}{425}}{\frac{1}{13} \times \frac{1}{425} + \frac{12}{13} \times \frac{2}{425}}$$

$$= \frac{1}{1 + 24} = \frac{1}{25}$$

Answer: $\frac{1}{25}$.